Design for Social Innovation

Functional Scope Document: Sprint 0 Recap

Project: 01B: Finding the Next Best Step
Team: DSIders

1. Problem & Context

- **Problem Statement:** Individuals from underserved, low-income backgrounds who have a clear aspiration struggle to identify their immediate next concrete step. The required information is scattered across fragmented, disconnected sources, and existing systems fail to help them identify what they do not know about their current situation (such as missing certificates or eligibility hurdles).
- **Existing Solution Gaps & Why They Fail:**
    - Static Government Portals (e.g., MyScheme, Jan Samarth): Require users to complete long demographic forms upfront or search using precise administrative vocabulary. They function as flat catalogs and perform binary filtering, disqualifying a user outright instead of resolving missing credentials.
    - Standard Conversational LLMs: Produce generic narrative advice, hallucinate scheme eligibility criteria, and lack deterministic graph traversals to ensure legal accuracy.
    - Manual NGO Casework: Highly accurate and empathetic, but geographically bounded, unscalable, and prone to counselor fatigue.
    - Counseling Deficit: More than 90% of Indian secondary and higher education institutions have no dedicated career counseling or guidance infrastructure.
    - Intermediation Gap**:** Organizations like the Barabari Collective and Nirmaan have to deploy manual mentorship and training workflows to bridge this structural divide.

2. User, Stakeholder & Environmental Map

User & Stakeholder Profiles

- **Primary Beneficiary:** Low-income youth, rural job seekers, women, and backward-class citizens seeking skill development or welfare schemes without direct access to counseling or administrative experts.
- **Secondary Stakeholder (Field Facilitators):** NGO caseworkers, village-level entrepreneurs (VLEs), and digital center operators (e.g., CSC, MeeSeva) who need high-throughput tools to triage candidates and diagnose blockers quickly.
- **Tertiary Stakeholder:** Government administrative departments and policy implementing bodies interested in higher conversion rates of allocated welfare budgets.

Operating Environment & Technical Constraints

- **Device Access:** Predominantly shared, budget Android smartphones (low RAM/storage) operating on intermittent 3G/4G bandwidth.
- **Linguistic & Cognitive Bandwidth:** Limited technical and legal English literacy; complex bureaucratic language causes drop-offs. Modality must support regional Indic vernacular and simplified input.
- **Document Physicality:** Supporting documents are often un-digitized, expired, damaged, or stored in native villages, making full documentation upfront impossible.

3. Key Pain Points & User Constraints

- **Missing Translation Layer:** Users do not suffer from a lack of aspiration; they lack a mechanism to decompose high-level intentions into actionable first steps.
- **The "Unknown Unknowns" Barrier:** Users often do not know what prerequisites they lack (e.g., specific income certificates, domicile papers, or registration deadlines) before an option can even be considered viable.
- **Cognitive Overload & Drop-off:** Standard interfaces force users through exhaustive, multi-field application forms or dump flat directories of maybe-relevant programs, stalling user momentum indefinitely.
- **Binary Disqualification Fallacy:** Traditional search engines treat missing data as disqualifying, whereas real-world pathways simply require an intermediate administrative step to resolve the dependency

4. Core Solution Concepts & Architectural Approaches

- **Smart Question Picker (Information-Gain Minimization):** Rather than long intake forms, an LLM extracts slot-filled attributes from free-text

goals, followed by an information-gain scoring algorithm that prompts one high-impact question at a time to prune candidate schemes rapidly.
- **Confidence-Tiered Bucketing:** Eliminates flat ranked lists in favor of three explicit buckets with explanations: *Confirmed*, *Likely (Pending Verification)*, and *Ruled Out*.
- **Prerequisite Chain Traversal:** A knowledge graph (Directed Acyclic Graph) maps relationships among goals, prerequisite certificates, schemes, and documents. Graph traversal algorithms locate and return the nearest unmet node (the immediate next dependency) rather than only pointing to the distant end goal.

5. UI/UX Interaction Context

- **Input Stage:** Single conversational prompt (text or voice) supporting natural goal phrasing without bureaucratic jargon.
- **Dynamic Elicitation:** Conversational single-question cards with binary or multi-choice tap targets, minimizing typing.
- **Visual Pathway Stepper:** An interactive vertical progression displaying the target aspiration at the top, with immediate prerequisite tasks pinned as physical milestones below.
- **Dual-View Switch:**
    - *Citizen Mode:* Vernacular cards with actionable instructions (e.g., *"Take these 2 documents to MeeSeva center"*).
    - *Caseworker Mode:* Compact checklist with manual override controls for NGO field operations.

6. Technology Stack & Integrations

- **Frontend & UX:** Figma for interactive prototyping; Next.js / React (mobile-first, low-bandwidth UI with accessible tap targets).
- **Indic Voice Layer:** Bhashini APIs (ASR for speech input, NMT for regional translation, TTS for read-aloud guidance).
- **AI & Extraction:** Quantized Small Language Model (SLM) via Ollama with Instructor/Outlines for structured JSON slot-filling from unstructured text.
- **Graph & Decision Engine:** Python with NetworkX (DAG modeling for prerequisite traversal) and NumPy (entropy-based active question picker).
- **Backend & Data:** FastAPI for REST orchestration; PostgreSQL/JSONB for curated scheme dependency rules and session state.

7. Open Questions & Assumptions to Validate

- **Scheme Rule Codification Ceiling:** To what degree can unstructured, variable government scheme guidelines (from sources like MyScheme) be reliably converted into automated, machine-checkable rules without requiring prohibitive manual curation?
- **Fidelity of Synthetic Data:** Whether synthetic test profiles can accurately capture real-world messiness, ambiguity, and conflicting user documentation. Validation must be performed directly against partner NGO feedback rather than internal assumptions.
- **Definition of Success:** A successful interaction means the user leaves with exactly one unambiguous next action (or a focused, short list of missing pieces needed to determine one), evaluated via the average number of questions asked before reaching a confident tiering without false positives.

8. Product Outcomes & What Works for Users

- **Clear, Actionable Next Steps:** The product directs users to their immediate next task—like visiting a local center for a specific missing certificate—rather than overwhelming them with long, complex eligibility catalogs.
- **Reframing Roadblocks as Milestones:** Missing criteria are shown as solvable prerequisites instead of outright rejections, keeping users motivated.
- **Reduced Caseworker Burden:** Field facilitators can assess eligibility quickly without manual scheme cross-referencing, freeing up time for in-person support.
- **Progressive, Single-Question Flow:** Asking one simple question at a time prevents the drop-off caused by lengthy static forms.
- **Visual Pathways Over Flat Lists:** Guided, step-by-step progressions work better for users than dense ranked lists, clearly demystifying what needs to be done next.

9. Out of Scope (Sprint 0 Boundary Setting):

- **Exhaustive National Ingestion:** Parsing all thousands of nationwide schemes; the prototype will be constrained to a tightly curated corpus of skill development and social welfare schemes.